OPERATING SYSTEM (OS)

UNIT-WISE IMPORTANT QUESTIONS

Selected from analysis of the three provided previous-year papers and your syllabus

★ = Higher-priority question

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UNIT 1: Overview of Operating System

1. ★ Explain different types of Operating Systems with examples.
2. ★ Explain the functions and services of an Operating System.
3. ★ What is a system call? Explain different types of system calls with examples.
4. ★ Explain the structure and architecture of UNIX/Linux with a diagram.
5. ★ What is a kernel? Explain its functions and services.
6. Explain system programs and their role in an Operating System.

UNIT 2: Process Management

1. ★ Explain different CPU scheduling algorithms with examples.
2. ★ Explain FCFS scheduling with a numerical example. Calculate average waiting time and turnaround time.
3. ★ What is a deadlock? Explain its four necessary conditions and methods of handling deadlocks.
4. ★ Explain deadlock prevention, avoidance and detection.
5. ★ What is Inter-Process Communication (IPC)? Explain its methods.
6. Explain process synchronization and the critical-section problem.

UNIT 3: Memory Management

1. ★ What is paging? Explain the paging mechanism with a suitable diagram.
2. ★ Explain virtual memory and page faults in detail.
3. ★ Explain First Fit, Best Fit and Worst Fit memory allocation techniques with examples.
4. ★ Differentiate between internal and external fragmentation. Explain compaction.
5. ★ Explain segmentation and differentiate between paging and segmentation.
6. Explain page replacement algorithms with examples.

UNIT 4: I/O System

1. ★ What are disk scheduling algorithms? Explain FCFS, SSTF, SCAN and C-SCAN.
2. ★ Explain disk structure and different methods of disk attachment.
3. ★ Explain RAID and its different levels with examples.

4. Explain swap-space management and its importance in an Operating System.
5. Explain buffering, caching and spooling with examples.
6. Explain the overview and functions of mass-storage management.

UNIT 5: File Management

1. ★ What is a file system? Explain its structure and implementation.
2. ★ Explain different directory structures with suitable diagrams.
3. ★ Explain file sharing and protection mechanisms in an Operating System.
4. ★ Explain different file allocation methods with their advantages and disadvantages.
5. Explain different file access methods with examples.
6. Explain free-space management techniques and different types of file systems.

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